

Original Manuscript ID: Access-2025-51751

Original Article Title: “Performance Indicators for Shape and Position Assessment in Electromagnetic Inverse Scattering”

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers’ comments.

We are uploading:

- (a) our point-by-point response to the comments (below) (response to reviewers, under “Author’s Response Files”),
- (b) an updated manuscript with yellow highlighting indicating changes (as “Highlighted PDF”), and
- (c) a clean updated manuscript without highlights (“Main Manuscript”).

Best regards,

Batista & Adriano.

Reviewer #1

Concern #1: *It is not sufficient to only report the values of the proposed metric. For a fair assessment, the authors should evaluate the same dataset using commonly adopted metrics such as MSE and SSIM, include metrics that assess contrast, such as CNR, and include shape and localization accuracy metrics such as the Dice coefficient, IoU (Intersection over Union), and the Hausdorff distance. A side-by-side comparison on identical data is necessary so that readers can clearly see in which situations the proposed metric is more sensitive or more informative than existing metrics. How useful is the proposed metric compared to conventional metrics? On the same dataset, please compute and report standard metrics such as MSE, SSIM, CNR, Dice, IoU, and Hausdorff distance together with the proposed metric. Analyze how each metric responds to variations in restoration quality. In particular, if there are cases where traditional metrics fail to distinguish differences in quality but the proposed metric can, please highlight such case studies. These examples would make the advantages of the proposed metric much more convincing.*

Author response: Insert your response here.

Author action: We updated the manuscript by...

Concern #2: *Especially in medical imaging, where clinical interpretation is critical, it is important to show how well the proposed metric values correlate with subjective assessments by clinical experts (e.g., physicians, radiologists, or other experienced observers). At present, the manuscript lacks such expert-based or clinically oriented validation. Including expert scoring of image quality or restoration adequacy, and analyzing its correlation with the proposed metric, would substantially strengthen the work. If the value of the proposed metric is high, can we truly say that the image has been accurately reconstructed? Please provide clear theoretical and/or empirical justification for this claim. For example, we recommend including analyses that show the correlation between the proposed metric and expert visual scores, demonstrating that the metric is consistent with clinical or expert judgment.*

Author response: Insert your response here.

Author action: We updated the manuscript by...

Reviewer #2

Concern #1: *The data source gives 3D phantoms (full 3D models), but the scattering model is in 2D. It would be useful to briefly explain what limitations arise when a 3D structure is simplified to 2D.*

Author response: Insert your response here.

Author action: We updated the manuscript by...

Concern #2: *When the original objects have a few pixels (i.e., small targets), the shape metric divides by a very small number (i.e., TP), so the error percentage becomes huge and unstable. A different shape metric (e.g., normalized, etc.) could address this issue.*

Author response: Insert your response here.

Author action: We updated the manuscript by...

Concern #3: *Please list here the third concern of reviewer 2...*

Author response: Insert your response here.

Author action: We updated the manuscript by...

Note: References suggested by reviewers should only be added if it is relevant to the article and makes it more complete. Excessive cases of recommending non-relevant articles should be reported to ieeeaccess@ieee.org.